Project Report

Dandruff Biomedical Knowledge Graph Construction Using Python and NetworkX

Submitted To:

Dr. Muahammad Ishaq

(Assistant Professor)

Submitted By:

Abdu Samad Shah

MS(DS) 2024-2026

SCHOOL OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

INSTITUTE OF MANAGEMENT SCIENCES PESHAWAR

March 2026

1. Introduction

Biomedical information can be widely dispersed in research articles, structured databases, and biological repositories It complicates the process of gathering information and comprehending it well. (Chandak et al 2023). Separately stored information can be easily overlooked thus causing one to lose the good links between diseases, microorganisms, drugs and biological processes (Santangelo et al 2025). This issue can be addressed by a knowledge graph, which models biomedical facts as networks and links, simplifying the exploration, validation, and interpretation of the information.

In this project, a disease-centered biomedical knowledge graph was created to structure the information connected with dandruff and seborrheic dermatitis. The graph is a synthesis of PubMed, PubChem and NCBI evidence. This method enables connecting microorganisms, disease states, and treatment choices into one network that is more or less connected, similar to the modern biomedical knowledge networks (Nicholson and Greene 2020). The project is based on a systematic development scheme, which comprises of multi-source data gathering, data synthesis, Python preprocessing, core relation filtering, and knowledge graph development with NetworkX.

Its key goal is to build a clean and significant biomedical network that may facilitate interpretation, visualization, and subsequent enhancement. Rather than storing the information separated in the rows of a spreadsheet, the project transforms it into a graphical form where nodes are biomedical entities and the connections between them are an edge. (Nicholson and Greene 2020).

2. Problem Statement

Biomedical information pertaining to scalp diseases like dandruff is spread across numerous sources and it is not always easy to interpret the information in one cohesive manner. The identical body can be presented in slightly altered format and relationships between diseases, microbes and drugs can be disjointed. The difficulty that this project aims to solve is to incorporate biomedical facts into a coherent and validated knowledge graph that illustratively demonstrates how entities are related to each other.

3. Objectives of the Study

The project was planned with the following goals:

1. To gather biomedical facts regarding dandruff and other scalp diseases in PubMed, PubChem and NCBI.

2. To clean and normalize names of entities and relation labels.

3. To build directed knowledge graph with NetworkX.

4. To sift the data and keep only useful core biomedical relations.

5. To derive disease centered, microbiome centered and drug centered subgraphs to be analyzed.

6. To authenticate the graph with sources metadata and confidence scores.

4. Data Sources and Dataset Description

The resulting processed dataset consists of biomedical relations that have been curated and obtained via various sources. The structure of the dataset includes the following fields:

- Subject
- Subject_Type
- Relationship
- Object
- Object_Type
- Source_Database
- Source_ID
- Evidence_Text
- Evidence_Type
- Confidence_Score

The database is small yet data-filled. It comprises a combination of microorganisms, drugs, chemicals, enzymes, substances, diseases, symptoms and biological processes. Your cleaned workbook constructed a graph out of 30 curated relations, having confidence scores between 0.70 and 0.95 and a mean confidence score of circa 0.86.

The major types of entities are:

- Microorganism
- Drug
- Chemical
- Enzyme
- Substance

The principal types of objects are:

- Disease
- Biological Process
- Symptom
- Chemical
- Enzyme
- Biological Structure
- Process
- Substance

Such structure ensures that the dataset is convenient to use in the construction of knowledge graphs since entity labels, relation labels, and provenance are already present.

5. Methodology

5.1 Study Design

The personal approach of this project is the applied graph-construction. This paper focuses on converting structured biomedical data into a directed knowledge graph that can be explored and proved.

5.2 Data Integration

All the data in PubMed, PubChem, and NCBI were integrated into one spreadsheet-based dataset. This move was done to make sure the graph was not anchored on one source, but on a number of complementary biomedical repositories. This is what is stated in the presentation file as organized data combination and multi-source data gathering.

5.3 Cleaning and Standardization

The text fields were cleansed before the graph was constructed to minimize formatting errors. This included:

- trimming extra spaces,
- standardize capitalization,
- normalize to lowercase the relation names.

This step helped ensure that the same entity was not treated as different nodes because of simple text variations.

5.4 Core Relation Filtering

Not all the relations present in the dataset were included in the final graph. The work concentrated on the fundamental biomedical relationships like:

- causes
- associated_with
- treats
- produces
- activates
- inhibits
- breaks_down
- damages
- protects
- decreases_in
- disrupts

This step of filtering was significant as it minimized noise and made the graph focused on biologically significant relationships.

5.5 Disease-Centered Filtering

The main disease nodes were then considered as the center of the graph:

- Dandruff
- Seborrheic Dermatitis

Extraction of relations between these diseases was in a way that produces a final network that would portray biomedical patterns of diseases in a clear manner.

5.6 Microbiome and Drug Subgraph Extraction

In order to analyze the structure of the network better, two other subgraphs were constructed:

- a network of microbiomes, which includes Malassezia, Staphylococcus, Streptococcus, Candida, Aspergillus, Cutibacterium and Filobasidium
- a drug network, holding treatment related entities, like Ketoconazole, Zinc Pyrithione, Selenium Sulfide, Fluconazole, Itraconazole and Ciclopirox.

Such subgraphs were useful in isolating mechanism of disease and relationships of treatment and maintaining them in the same biomedical graph.

5.7 Graph Construction in NetworkX

NetworkX was used to make a directed graph. In this graph:

- Biomedical entities are represented by nodes,
- Relation is represent by edges,
- Entity type of the node is store as attribute,
- edge attributes store relation type, source database, source identifier, evidence text, and confidence score.

This design renders the graph understandable and identifiable.

5.8 Validation

The project will involve source and structural validation. In the presentation, it is stressed that fact checking was done with PubMed, PubChem, and NCBI, and standardized naming minimized repetitive and conflicting entries. The reliability of relations was also estimated by measuring confidence scores.

6. Results

6.1 Dataset Summary

The filtered dataset was transformed to a knowledge graph dataset of curated biomedical relations in a compact format. The data was preprocessed, and the structured format of the data in the form of a graph, along with the source information and confidence values, was obtained.

6.2 Main Graph Insights

The graph indicates that the disease nodes are the central nodes in the biomedical network. Dandruff and seborrheic dermatitis is related to microorganisms, biochemical processes, and treatment drugs. It is apparent that it has a disease-centered organization as opposed to a plain list of facts.

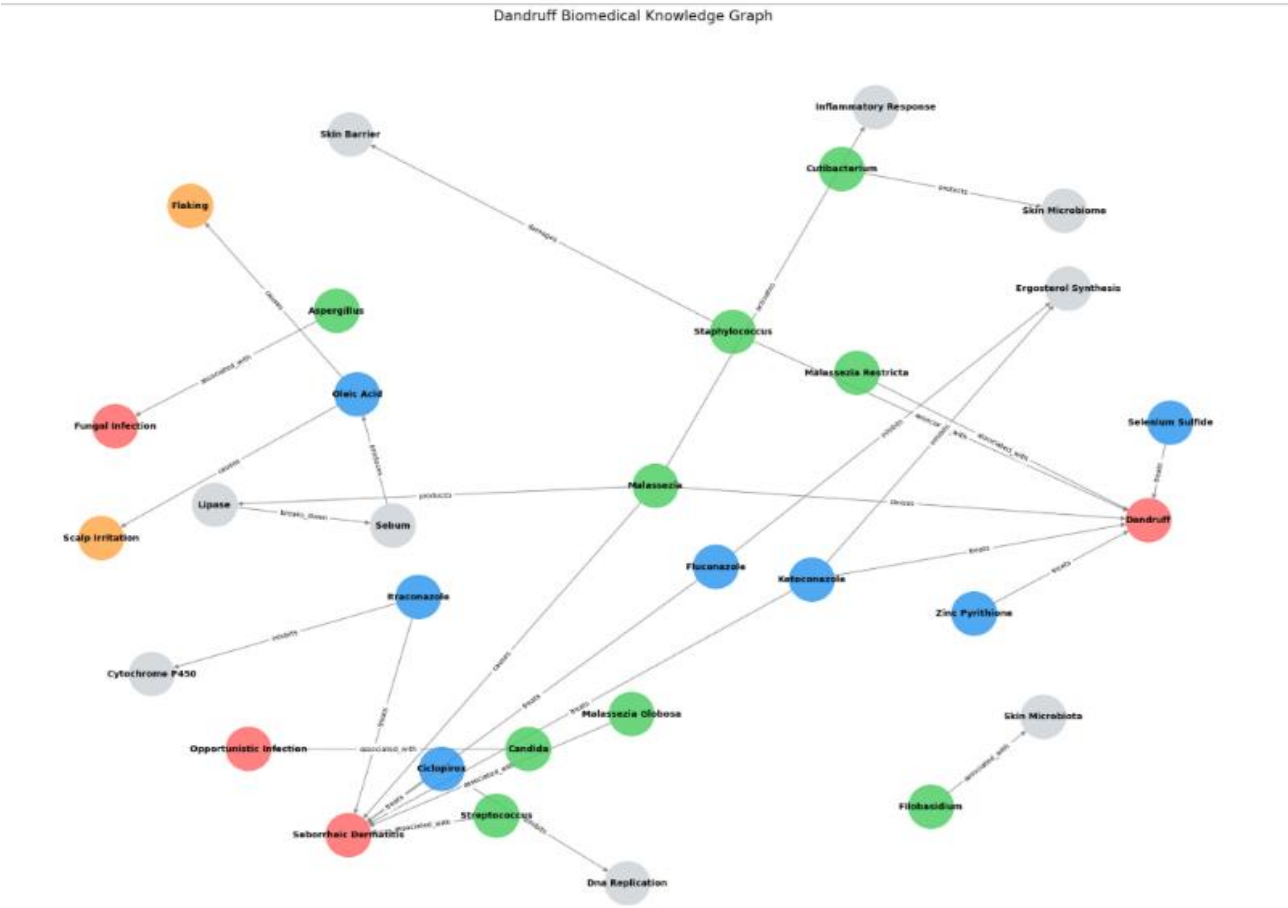

Figure 1. Dandruff Biomedical Knowledge Graph.

The crucial trends that can be observed in the graph are:

- Malassezia has been found as an important microorganism associated with mechanisms of scalp-related disease.
- There are a few microbes in the same disease scenario that indicate that dandruff is not linked with a single biological agent
- There are treatment relations between drug nodes and disease nodes.
- The graph indicates an understandable sequence of pathways like microorganism - disease-drug.

6.3 Relation Patterns

Three key knowledge directions are evident in the curated relations:

1. Association and causation of disease
2. Metabolic activity and involvement in biology.
3. Mechanisms of drug treatment and inhibition.

The graph is applicable in biomedical interpretation due to these patterns that relate pathology to mechanism and treatment.

6.4 Structural Strength of the Graph

The graph is not simply a list of facts but it is an interconnected biomedical network. Since provenance and confidence information are stored at each edge, the graph can be explored and added to in the future with additional sources or additional diseases.

7. Discussion

The project shows that when biomedical knowledge is presented as a graph instead of being in discrete rows in a spreadsheet as we will talk about later that knowledge graph help us visualize how various things are related and easier to explore complex biological systems in a structured way (Wang et al. 2025).

Dandruff and seborrheic dermatitis are not used as individual terms in the last network. They instead serve as hubs that are linked to microorganisms, biological processes, and treatment alternatives.

The structure is useful as it can be used to analyse a variety of types:

- following the relationship between microorganisms and scalp diseases,
- identifying treatment links,
- comparing source reliability,
- and searching for latent associations among biological entities.

It was also demonstrated in the work that not all relations are to be included automatically. The ultimate graph is narrow and significant, and we eliminated all but the core relations to ensure that the data is clean and meaningful since Earlier research also indicates that biomedical knowledge graphs should be planned, have consistent standards, and have good data curation to eliminate noise and make them more reliable and useful (Cortes et al 2025). The knowledge graph is further enhanced by the use of source metadata and confidence scores.

On the whole, the project demonstrates that a small yet well-selected biomedical dataset may yield a powerful and explainable graph provided the process of data cleaning, relation selection, and graph-building is performed correctly. This is in line with previous studies, which indicate that even smaller, curated knowledge graphs can still be useful in biomedical analysis and assist in discovering valuable insights (Zhang et al 2025)

8. Conclusion

This initiative was able to create a disease-centered biomedical knowledge graph of dandruff and related scalp diseases using Python and NetworkX. PubMed, PubChem and NCBI biomedical information were gathered, cleaned, standardized, filtered, and transformed into a graph format.

The last graph emphasizes the key functions of microorganisms, illness states, and antifungal drugs in the sphere of dandruff. It also gives a ready and adaptable framework to expand in the future. More biomedical entities, diseases or drug interactions can be included later to render the graph more complete.

Conclusively, this project illustrates that the concept of knowledge graphs provides an effective means of managing biomedical knowledge in a manner that is interconnected, comprehensible and expandable.

9. Limitations and Future Work

The existing graph is specialized to a small disease-based domain thus restricted to the curated entities and relations that have been extracted previously. To enhance the project in future work, the project can be enhanced by:

- to more diseases and symptoms,
- introducing additional types of relations,
- incorporating ontology-based normalization,
- enhancing automatic literature harvesting,
- and representing the graph with the interactive tools.

Such enhancements would result in the knowledge graph being more comprehensive and more beneficial to biomedical exploration.

References:

1. Chandak P, Huang K, Zitnik M. Building a knowledge graph to enable precision medicine. Scientific data. 2023 Feb 2;10(1):67.

2. Nicholson DN, Greene CS. Constructing knowledge graphs and their biomedical applications. Computational and structural biotechnology journal. 2020 Jan 1;18:1414-28.

3. Santangelo BE, Bada M, Hunter LE, Lozupone C. Hypothesizing mechanistic links between microbes and disease using knowledge graphs. Scientific Reports. 2025 Feb 26;15(1):6905.

4. Su C. MedRule-KG: A Knowledge-Graph--Steered Scaffold for Reliable Mathematical and Biomedical Reasoning. arXiv preprint arXiv:2511.12963. 2025 Nov 17

5. Wang L, Hao H, Yan X, Zhou TH, Ryu KH. From biomedical knowledge graph construction to semantic querying: a comprehensive approach. Scientific Reports. 2025 Mar 12;15(1):8523.

6. Cortes KG, Sundar S, Gehrke S, Manpearl K, Lin J, Korn DR, Caufield H, Schaper K, Reese J, Koirala K, Hunter LE. Improving biomedical knowledge graph quality: A community approach. ArXiv. 2025 Aug 29:arXiv-2508.

7. Zhang Y, Sui X, Pan F, Yu K, Li K, Tian S, Erdengasileng A, Han Q, Wang W, Wang J, Wang J. A comprehensive large-scale biomedical knowledge graph for AI-powered data-driven biomedical research. Nature Machine Intelligence. 2025 Apr;7(4):602-14.